

Kaichen Huang 黄楷宸

Ph.D. Student, College of Computing and Data Science, Nanyang Technological University

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Education

Nanyang Technological University — College of Computing and Data Science 2025.08 – Present

Ph.D. Student · Advisor: Prof. Bo An

Nanjing University — School of Artificial Intelligence

2022.09 – 2025.06

M.Sc. (GPA 4.56/5.0) · LAMDA Group · Advisor: Prof. De-Chuan Zhan

Nanjing University — School of Artificial Intelligence

2018.09 – 2022.06

B.Sc.

Research Interests

World models and interactive / streaming video generation; reinforcement and imitation learning; interpretability of multimodal large language models.

Publications

* Equal contribution.

[1] Runjia Qian*, Zile Wang*, Jihai Zhang*, Kai Zou*, Wei Yu*, Jiaying Li*, Zexiang Liu*, Yaokun Li, Fei Kang, **Kaichen Huang**, Mengyin An, et al.. *Matrix-Game 3.5: Enhancing Real-Time Streaming Interactive World Models with Patch Memory*. [Preprint, 2026](#).

[2] Jiaying Li*, Kai Zou*, Cindy Zhou, **Kaichen Huang**, Junyao Gao, Zile Wang, Yang Liu, Bin Liu, Bo An, Yangguang Li. *DistillAlign: Coordinating Mode Covering and Mode Seeking in Autoregressive Video Distillation*. [Preprint, 2026](#).

[3] Zile Wang*, Zexiang Liu*, Jiaying Li*, **Kaichen Huang***, Baixin Xu*, Fei Kang, Mengyin An, et al., Yangguang Li. *Matrix-Game 3.0: Real-Time and Streaming Interactive World Model with Long-Horizon Memory*. [Preprint, 2026](#).

[4] Jiaying Li*, **Kaichen Huang***, Baixin Xu*, Zexiang Liu, Xianglong He, Zile Wang, Junyao Gao, Yang Liu, Ying He, Bo An, Yangguang Li. *Direct Autoregressive Diffusion Distillation via Error-Aware Causal Pretraining*. [ECCV, 2026](#).

[5] Wenshu Fan, Shenghua Wan, Xinchun Li, Hai-Hang Sun, **Kaichen Huang**, Le Gan, De-Chuan Zhan. *Behavior Cloning Optimization via Secondary Learning*. [CCML, 2025](#).

[6] Shanghai AI Lab (**Kaichen Huang** as contributor). *SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law*. [Preprint, 2025](#).

[7] Yunkai Dang*, **Kaichen Huang***, Jiahao Huo*, Yibo Yan, Sirui Huang, Dongrui Liu, Mengxi Gao, Jie Zhang, Chen Qian, Kun Wang, Yong Liu, Jing Shao, Hui Xiong, Xuming Hu. *Explainable and Interpretable Multimodal Large Language Models: A Comprehensive Survey*. [Under review, TPAMI](#).

[8] **Kaichen Huang***, Shenghua Wan*, Minghao Shao, Shuai Feng, Le Gan, De-Chuan Zhan. *Leveraging Separated World Model for Exploration in Visually Distracted Environments*. [NeurIPS, 2024](#).

[9] **Kaichen Huang**, Jiahao Huo, Yibo Yan, Kun Wang, Yutao Yue, Xuming Hu. *MINER: Mining the Underlying Pattern of Modality-Specific Neurons in Multimodal Large Language Models*. [Preprint, 2024 \(submitted to ICLR 2025\)](#).

[10] **Kaichen Huang***, Minghao Shao*, Shenghua Wan, Hai-Hang Sun, Shuai Feng, Le Gan, De-Chuan Zhan.

SENSOR: Imitate Third-Person Expert's Behaviors via Active Sensoring. [Preprint, 2024](#).

[11] **Kaichen Huang**^{*}, Hai-Hang Sun^{*}, Shenghua Wan, Minghao Shao, Shuai Feng, Le Gan, De-Chuan Zhan. *DIDA: Denoised Imitation Learning based on Domain Adaptation*. [Preprint, 2024](#).

[12] Yihan Liu, **Kaichen Huang**, Yechao Bai. *An Error Analysis Method for Externally Measured Data*. [Measurement and Control Technology, 2023](#).

Experience

Skywork AI · Riemann Dynamics

2025.08 – 2026.10

Joint-Training Ph.D. Student · Mentors: Yangguang Li, Wei Yu

Real-time interactive world models and video generation — Matrix-Game 3.0 / 3.5, DistillAlign, Error Forcing (ECCV 2026).

Shanghai AI Laboratory

2024.11 – 2025.03

Research Intern · Mentor: Jing Shao

Multimodal safety and reasoning — SafeWork-R1.

Hong Kong University of Science and Technology (Guangzhou)

2024.06 – 2024.09

Research Assistant · Mentor: Xuming Hu

Interpretability of multimodal LLMs — MINER, and a survey on explainable & interpretable MLLMs.

Ministry of Industry and Information Technology (MIIT)

2023.06 – 2023.12

Data Processing Engineer

Automated analysis system for industrial economic-trend data; resulted in a granted software patent.

Pengcheng Laboratory

2022.06 – 2023.03

Assistant Engineer · Mentor: Tong Zhang

Medical imaging for congenital heart disease — a ViT / MAE-based ellipse-detection method for robust fetal echocardiography measurement.

Honors & Awards

2023 Academic Scholarship, Nanjing University

2020 First Prize (East China Region), National Undergraduate IoT Design Competition (Huawei Cup)

2019 Third Prize, 9th Nanjing University College Student Career Planning Competition

Academic Service

Conference Reviewer: NeurIPS, ICML, ICLR, AAAI.

Language & Test Scores

IELTS: Overall 7.0 (L 7.5 / R 8.5 / S 5.5 / W 6.5) · GRE: 321 (V 151 / Q 170 / AW 4.5)